PART I · THE ARGUMENT

A Science and Its Discipline

OPENING CASE

The discipline that changed its mission

DATED: FEBRUARY 2026. FOUND BY SEARCH. FIGURES AND THE QUOTATION AGREE ACROSS SECONDARY COVERAGE AND HAVE NOT YET BEEN CHECKED AGAINST THE REPORT ITSELF. SEE THE SOURCE REGISTER.

In February 2026, the FinOps Foundation revised its mission statement. For years, it said it existed to advance the people who manage cloud value. The new statement changed a single word: it now advances the people who manage technology value.[1]

The change accompanied the Foundation's annual survey of practitioners: people inside companies who manage cloud spending. The 2026 survey included 1,192 respondents who collectively oversaw more than $83 billion in annual cloud spending.[2] Its central finding concerned work the discipline was not originally created to do. Ninety-eight percent of respondents now managed AI spending, up from 31 percent two years earlier.[3]

FinOps was built to solve a specific problem: engineers could buy cloud capacity with a few lines of code, but the bill arrived only after they had used it. The discipline emerged to make engineering, finance, and business teams jointly accountable for that spending.[4] Its practitioners

1. FinOps Foundation, "State of FinOps Survey: AI Value and Skills Top Priorities as FinOps Matures Across Technology Value," February 19, 2026, accessed October 1, 2026. The mission statement before and after the change.

2. FinOps Foundation, "State of FinOps 2026," February 19, 2026, accessed October 1, 2026. Survey size and the cloud spending its respondents manage.

3. FinOps Foundation, "State of FinOps 2026," February 19, 2026, accessed October 1, 2026. Share of practitioners managing AI spend, 2026 against 2024.

4. FinOps Foundation, "What is FinOps?," accessed October 1, 2026. The practice as the Foundation defines it.

later took on AI spending as well, and by 2026, nearly all of them managed it.

The report also recorded a question this work had not yet answered. Asked whether AI was delivering value, one practitioner replied in words the report chose to quote: “No one can answer that question yet.”[5]

The survey reveals the gap Chapters 1 and 2 described among the very people expected to close it. For an organization using a metered resource, cost arrives by default with the bill, while value becomes visible only when someone builds a way to measure it. Cloud-cost management is a neighboring discipline, but it has been drawn, reluctantly, into work it was never designed to do: managing the economics of an organization’s AI use.

FinOps can show who is using AI and what the organization pays for it. On its own, however, it cannot determine whether the system produces accurate, useful work or build the management practice that turns that evidence into decisions. That work requires both a body of knowledge for judging what is true and a practice for acting on those judgments. Chapter 14 explains how FinOps and that practice divide the work. Until both exist, the practitioner’s question about whether AI provides value remains unanswered.

3.1 Why name a science

Managers act on claims about how their business works. A budget assumes that cost follows usage, a pricing negotiation assumes the provider has a reason to accept a lower price, and a dashboard assumes that the information it shows will change someone’s actions. When those claims are wrong, the management built on them fails, however carefully managers carry it out.

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5. FinOps Foundation, “State of FinOps 2026,” February 19, 2026, accessed October 1, 2026. Practitioner quotation as reproduced in the report.

These claims can remain informal, held in experience, rules of thumb, or the slides of whoever presented last. Informal knowledge is useful, but it has a cost: two managers can hold opposite beliefs about the same deployment without being able to identify where they disagree. One might treat Chapter 1's Cursor licenses as a fixed software expense and the other as metered consumption. Unless each states the claim behind that view, neither can identify the fact that would settle the disagreement, and the argument turns on confidence rather than evidence.

Chapters 1 and 2 each ended with a stated result rather than an opinion. Chapter 1 set out the conditions under which deployed AI is a resource the organization consumes, while Chapter 2 set out the conditions under which a scaled deployment cannot be controlled without a governing apparatus. Each result appeared in a panel with an identifier that points to the larger body of claims supporting it.

That body of claims is AI Business Economics. It is a science in a specific and limited sense: a set of claims about the business economics of consuming AI, each stating when it holds and precise enough to be checked. Here, science means neither laboratory experiments nor certainty. It means that every claim states when it applies, what it concludes, and which other claims it rests on.

DEFINITION

AI Business Economics

The science of the business economics of consuming AI: an ordered body of conditional claims, each stating when it applies, what it concludes, and which other claims it depends on.

The claims are kept in a registry: a structured record of every definition, assumption, and result in the science, with each result linked to the claims that support it. A definition fixes what a term means, while an axiom states an assumption the science accepts without proof. A proposition states a single claim, accepted on the basis of the definitions, axioms, or earlier claims it names as support. A lemma combines propositions, and sometimes other lemmas, into an intermediate result. A theorem combines lemmas into a result with direct consequences for management.

Those links are the registry's most important feature. Every higher claim names the lower claims it depends on, which in turn name their own supporting claims. Together, the links form a dependency graph: a map in which no claim can rest, directly or indirectly, on itself. Anyone with the registry can start at a theorem and follow its supporting claims down to definitions and axioms. Appendix A reproduces every certified theorem and lemma, while the book cites propositions by identifier. Readers can therefore trace any claim to the lemma level unaided. Below that level, they work from the propositions supplied by a chapter or exercise.

For a manager, the graph rules out circular reasoning: no conclusion depends on itself. A manager who doubts a conclusion can therefore trace that doubt to a smaller claim and examine it. The graph does not, however, establish whether that smaller claim is true.

The registry also records which claims are complete. A claim is certified when every claim it depends on is itself certified and its supporting argument meets the registry's fixed conditions. The registry's validator computes that status from the graph at each version; no one enters it manually. The validator checks that every dependency is recorded and certified, that no chain of support loops back on itself, that every claim marked empirical has evidence attached, and that a reviewer has signed off on the claim's current wording.[6] These checks establish the structure and record of an argument, not its truth. If a claim has a gap in its support, the registry records what is missing. This book presents and cites only certified claims, because presenting an uncertified claim as a theorem would assert more than the science does.

The registry justifies this book, but it does not organize it. The chapters follow the order in which a manager needs to learn, which differs from the order in which the science derives its results. The registry supports each result the book teaches, allowing a reader who doubts a claim to trace its support. If the chapters instead followed the order of proof,

6. AI Business Economics registry, "dag.aiom, commit 9d7ee50," August 21, 2026. What the validator checks.

managers would spend many pages on definitions before reaching a decision they recognize.

3.2 Tracing a result the reader already accepts

Once a deployment reaches scale, economic control requires a governing apparatus; diligence from the people running it is no longer enough. Chapter 2 ended with that result, so a reader who accepted the chapter's argument already accepts the conclusion. Tracing that familiar result through the registry is the clearest way to learn how the registry works. The only new task is to understand how its supporting claims are arranged.

The registry states that result as THM-004:

> THEOREM 2 · THM-004
>
> **Scaled AI Deployment Requires Cost Governance for Economic Control**
>
> Within the defined deployment and measurement boundary, if:
>
> (i) production scaling expands the usage surface of AI deployment;
> (ii) measurement enables visibility into measurable cost-bearing activity;
> (iii) visibility enables management mechanisms; and
> (iv) buyer total cost extends beyond access price;
>
> then scaled AI deployment requires cost governance for economic control.

Each condition has a plain meaning a manager can check against a deployment. The first says that the deployment has moved from a pilot into daily operations, multiplying the places where usage can occur. The second says that measuring usage makes activity that costs money visible. The third says that managers can apply monitoring, limits, attribu-

tion, or other management mechanisms to visible activity. The fourth says that the organization's total cost exceeds the price of access. When all four hold, an organization seeking control over the deployment's economics needs cost governance.

THM-004 rests on four certified lemmas, one for each condition. LEM-016, "Production Scaling Expands Usage Surface," supports the first; LEM-002, "Measurement Enables Visibility," supports the second. LEM-020, "Visibility Enables Management Mechanisms," supports the third, and LEM-006, "Buyer Total Cost Extends Beyond Access Price," supports the fourth.

The registry also records the theorem's proof, which combines the four lemmas in two pairs. The first and fourth describe economic exposure that grows with the deployment and cannot be determined from the access price alone. The second and third describe the only route to managing that exposure: measurement makes it visible, allowing managers to apply management mechanisms. Control therefore requires an apparatus for both measurement and management, which the theorem calls cost governance.

This pairing explains Chapter 2's claim that THM-004 formalizes the economic consequence of the third flow's asymmetry. The first and fourth conditions describe cost that arrives by default and grows with scale. The second and third describe the measurement and management that an organization must deliberately build to control it.

One level down, each lemma rests on propositions. LEM-020 offers a useful example of how the registry is built. Its statement reads: "If economic exposure or usage activity is visible to the relevant actor, then monitoring, limits, attribution, or management mechanisms can be applied within the relevant scope." The lemma rests on seven propositions and one other lemma.

The propositions are small claims, each nearly obvious on its own. PROP-043 states: "If activity is metered, records describing measured activity are generated or maintained." PROP-053 states: "If economic

exposure exists, monitoring mechanisms may be used to observe the sources or magnitude of that exposure." PROP-085 states: "If usage information becomes visible, decisions regarding management, allocation, optimization, or monitoring may reference that information." No manager would dispute these claims; the work lies in combining them. PROP-043 says that metered activity leaves a record, PROP-085 allows management decisions to draw on visible usage, and PROP-053 adds that monitoring can be used to observe economic exposure. None of the three alone establishes that controls can be applied to what is visible. LEM-020 reaches that conclusion by combining them with four further propositions and LEM-002, a combination the registry records as its proof.[7]

LEM-020 also rests on LEM-002, which directly supports THM-004's second condition. Figure 3.1 shows the branch through LEM-020 and makes that shared support visible. The registry is therefore a graph, in which one claim can support several others, rather than a tree, in which each claim has one parent.

7. AI Business Economics registry, "dag.aiom, commit 9d7ee50," August 21, 2026. LEM-020's proof record and its supporting propositions.

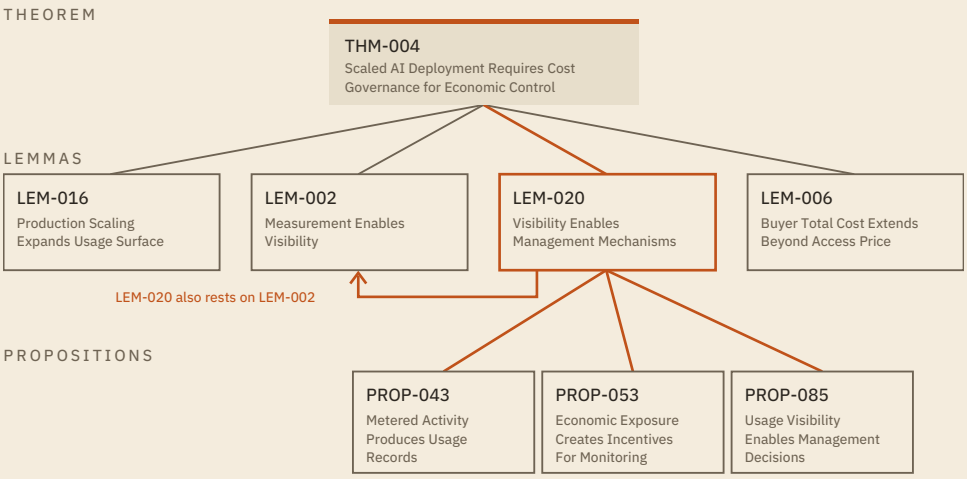


Figure 3.1 *One branch of the trace. THM-004 rests on four lemmas, one for each condition. LEM-020 rests in turn on LEM-002, which also supports the theorem directly, and on seven propositions, three of which are shown.*

Tracing the theorem changes how a manager can assess it. Before the trace, THM-004 is a conclusion to accept or reject as a whole. Afterward, the manager can examine its smaller supporting claims individually and identify which one they doubt. That gives the disagreement a point that can be checked: a dispute over an entire conclusion has no obvious resolution, but a dispute over one proposition can be settled.

> **DATED: AUGUST 2026**
>
> The registry version used to write this book held 413 objects: 140 definitions, 35 axioms, 201 propositions, 21 lemmas, 11 theorems, and 5 records of evidence the science still needs. It recorded 1,777 dependency links among them, and 397 of the 413 objects were certified.[8] These figures describe one version of the registry, which is revised as the science develops, rather than the science itself.

8. AI Business Economics registry, "dag.aiom, commit 9d7ee50," August 21, 2026. Object counts, links and certification at the pinned registry version.

3.3 How to read a formal claim

A manager who cites a formal claim for more than it says bases a decision on something the claim never established. To avoid that error, read the claim in four parts: its conditions, its scope, its non-claims, and the falsifying case that would defeat it. A manager who can state all four can use the claim without being misled by it.

The conditions are the "if" clauses, each of which is true or false in a given case. THM-004 has four numbered conditions, and if a deployment fails any one of them, the theorem says nothing about it. A pilot used by one team for a few weeks has not been scaled into daily operations, so it fails the first condition. The theorem therefore neither requires governance for the pilot nor rules it out. Reading the conditions first prevents a basic error: applying a formal claim where it was never stated to hold.

The scope defines which cases the claim addresses. As presented in this book, THM-004 states its scope before the word "if": one defined deployment and its measurement boundary. The registry adds that the theorem applies where an organization seeks economic control over the resulting cost.[9] The claim thus concerns one deployment within one boundary, rather than an organization's AI activity in general. To apply it across a whole company, a manager must first show that each deployment meets the conditions.

The non-claims identify what the claim deliberately does not establish. THM-004 says that governance is necessary for control, but it does not say that governance delivers control or is costless. It specifies neither the scale at which the requirement begins to apply nor the size of the exposure.[10] It also does not establish Chapter 2's three flows, which belong to this book's framework rather than the registry's. The registry

9. AI Business Economics registry, "dag.aiom, commit 9d7ee50," August 21, 2026. THM-004's scope condition.

10. AI Business Economics registry, "dag.aiom, commit 9d7ee50," August 21, 2026. THM-004's recorded exclusions.

records each claim's non-claims. A reader with only the statement can identify most of them by asking whether a requirement is sufficient or merely necessary, whether the claim specifies an amount, and what it says about cost.

DEFINITION

Non-claim

A conclusion a formal claim deliberately does not establish, though a reader might be tempted to draw it. Stating the non-claims prevents a claim from being cited for more than it says.

The most costly misreading confuses necessary with sufficient. A manager who reads THM-004 as "governance produces control" might buy a dashboard and expect costs to settle. The theorem makes no such promise: it says only that without governance, there is no route to control. One level down, LEM-020 makes the same limit explicit: it does not claim that management mechanisms are automatically effective.[11]

The falsifying case identifies what would show the claim to be wrong. For a conditional claim, the test is precise: a case that meets every condition but does not satisfy the conclusion. The second and third conditions describe what measurement and management would make possible, rather than an apparatus already in place, so a deployment with no instruments installed can still meet all four. For THM-004, the falsifying case would be a scaled deployment that meets all four conditions and whose organization controls its cost without any measurement and management apparatus. A flat-fee contract does not provide such a case. If the fee covers every cost, the fourth condition fails; if it does not, the organization would have to control the costs beyond the fee without measuring them. A claim that names no falsifying case cannot be wrong, and a claim that cannot be wrong tells a manager nothing.

These four parts answer an objection the formal style invites: that the registry consists of word games, dressing obvious points in numbered clauses. The common-sense version of THM-004 says that big deploy-

ments need cost controls, but it offers no answer for the one-team pilot above because it never defines how big is big. The formal version gives the manager a clear first task: check whether the deployment has moved into daily operations. The pilot fails that condition, so the theorem says nothing about it. The common-sense version also slides easily from "needs controls" to "controls deliver control," a misreading the non-claims rule out. A formal statement fixes in advance what would count as the claim being wrong. By stating its conditions, it makes the claim testable and therefore open to improvement.

3.4 The discipline that acts on the science

Knowing what is true does not, by itself, change what an organization does. Medicine depends on physiology, but clinicians must apply that knowledge to treat patients. Engineering depends on physics, but engineers must work to codes and procedures to build a bridge. In each case, a practice connects knowledge to results through its own methods, roles, and standards.

AI Operations Management is that practice for AI consumption: the work an organization does each week to manage what it consumes. The organization buys AI capacity, plans its use, records that use, and assigns it to the work it serves. It decides what runs when capacity is short and holds spending accountable for what it returns. Chapter 1 described the problem as a category error. AI Operations Management resolves that error by managing AI as a resource the organization consumes rather than as software it has bought.

The discipline has five functions that follow AI capacity through its life in the organization: buying it, planning its use, recording and assigning that use, routing capacity to work, and judging what it returns. Each function

DEFINITION

AI Operations Management

The management discipline that acts on AI Business Economics: the practice by which an

is a capability an organization either has or lacks. Part III teaches all five, with metering and attribution covered in two chapters.

organization buys, plans, records, assigns, allocates, and holds accountable its consumption of AI.

The first function, sourcing, matches the AI capacity an organization buys to the work it needs to do. The organization must avoid both paying for capability the work does not use and providing too little for work that needs more. Sourcing determines which models and providers serve which work, and on what terms. A company that sends every request to its most capable model, including requests a smaller model would answer just as well, pays for capability its work does not use.

The second function is planning and budgeting, whose absence proved costly in Chapter 2's opening case. The organization must be able to state how much AI it expects to consume in the coming period and recognize when actual consumption departs from that expectation. In this sense, a budget is a commitment made in advance that makes a deviation visible as it happens, rather than after the money is gone.

The third function, metering and attribution, builds what Chapter 2 called the record flow. The organization must record what was consumed and assign each unit of consumption to the team, product, or purpose it served. Without that assignment, managers can see a cost but cannot trace it to the work that caused it.

The fourth function is allocation and routing. When capacity is limited, the organization must decide what runs first and direct each piece of work to capacity that suits it. These decisions occur even when no one makes them deliberately. If two workflows share a provider account with a fixed rate limit, one waits whenever the other is busy. Allocation and routing make that choice a matter of rule rather than accident.

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11. AI Business Economics registry, "dag.aiom, commit 9d7ee50," August 21, 2026. LEM-020's recorded exclusions.

The fifth function is holding value to account. Consider a constructed example in which a support team adds an AI assistant to its help desk. At the end of a quarter, the organization should be able to say whether the assistant paid for itself by identifying what it cost, what it returned, the months covered, and the outcome used to judge it. The registry's LEM-021, "ROI Evaluation Requires Measurement Boundary," states the requirement: "If ROI is evaluated for AI adoption, the evaluation requires a defined boundary for relevant costs, value, time period, and observed or expected outcomes."[12] The organization must set that boundary for each deployment and identify who is responsible for knowing whether it paid for itself. This book calls it the value boundary.

Figure 3.2 shows the two layers together: the science forms the foundation, and the discipline builds on it through its five functions.

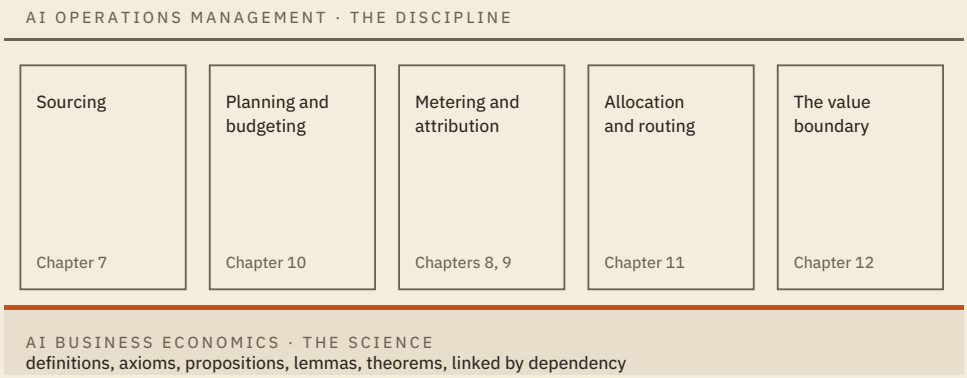


Figure 3.2 *The two layers. AI Business Economics states what is true about consuming AI. AI Operations Management acts on it through five functions, all taught in Part III.*

The two layers depend on each other in different ways. A result about governance changes nothing until an organization builds the apparatus it describes, so the science needs the discipline to put its results into practice. The discipline, in turn, needs the science because a practice built on wrong claims fails however well it is run.

12. AI Business Economics registry, "dag.aiom, commit 9d7ee50," August 21, 2026. LEM-021's statement.

3.5 Where the discipline ends

Chapter 1 set aside three subjects that lie close to AI but outside this book: how models work, prompt engineering, and finding uses for AI. Each has its own purpose and literature, and each ends before AI Operations Management begins.

The first subject is how models work, including their architectures, training, and internal behavior. It belongs to computer science and machine learning engineering and ends where a model becomes a service an organization can call. AI Operations Management begins with that call, where each use consumes resources and costs money.

The second subject is prompt engineering, the craft of writing inputs that produce better outputs. It improves individual uses of AI, while AI Operations Management governs those uses together: how many there are, what they cost, and what they return. A better prompt can lower the cost of one request, but only the discipline can say whether the organization's requests, taken together, are worth their cost.

The third subject is finding uses for AI: deciding which business problems it might solve. That work belongs to strategy and innovation and ends when a use is chosen. AI Operations Management begins with buying capacity for that use and continues to manage it thereafter.

Four established disciplines lie closer still. Each manages work adjacent to AI consumption, but each lacks something needed to govern its economics. Table 3.1 compares them.

NEIGHBOR	WHAT IT IS FOR	WHAT IT MANAGES	WHAT IT LACKS FOR THIS TERRITORY
AIOps	Applying machine learning to run IT operations: correlating events, detecting anomalies, finding causes[13]	The health and performance of IT systems	It uses AI to watch systems. It does not govern what consuming AI costs or returns
MLOps	Building, deploying, and monitoring machine learning models in production[14]	The lifecycle of models an organization builds	It ships models. It does not govern the consumption of capacity an organization buys
FinOps	Financial accountability for technology spending, through collaboration among engineering, finance, and business teams[15]	Cloud and wider technology spending	It was built around cloud billing. A practitioner told its own survey that no one could yet say whether AI provides value
Regulatory AI governance	Ensuring that uses of AI are lawful, safe, and accountable to regulators	Risk and compliance	It asks whether a use is permitted, not whether it is economically managed

Table 3.1. The four neighboring disciplines. Each manages something adjacent to AI consumption, and none was built to govern its economics.

FinOps is the nearest neighbor, making its border with AI Operations Management the hardest to draw. It already manages spending, has moved toward AI, and defines its purpose in terms of maximizing technology's business value.[16] Chapter 14 examines how the two divide the work. Their border rests on what each was built to do, rather than on

13. Gartner, "AIOps (Artificial Intelligence for IT Operations)," *Gartner Information Technology Glossary*, accessed October 1, 2026. The originating definition of AIOps.

14. Dominik Kreuzberger, Niklas Kühl, and Sebastian Hirschl, "Machine Learning Operations (MLOps): Overview, Definition, and Architecture," *IEEE Access* 11 (2023): 31866-31879. Definition and scope of MLOps.

15. FinOps Foundation, "What is FinOps?," accessed October 1, 2026. The Foundation's own definition of the practice.

16. FinOps Foundation, "What is FinOps?," accessed October 1, 2026. The Foundation's definition, which names business value.

intent: FinOps was built to make cloud bills accountable after capacity had been used. AI Operations Management manages consumption metered by the request and judges it against a value boundary set before spending begins.

Regulatory governance determines whether an organization may use AI in a given way. This book takes that answer as given and asks how the organization manages the permitted use.

3.6 Five questions the discipline exists to answer

AI Operations Management exists to answer five questions an organization cannot answer without the discipline. Each question belongs to one of the five functions and is answered in the chapter that teaches it. Part III presents the functions in a different order, for reasons explained there.

An organization that manages its AI consumption can answer these five questions from its own records:

Question 1. “Are you paying for capability you do not need, or starving work that needs more?” This is the sourcing question, answered in Chapter 7.

Question 2. “What do you expect the AI flow to consume next period, and how will you know when it deviates?” This is the planning and budgeting question, answered in Chapter 10.

Question 3. “Who consumed what last month, in service of which work?” This is the metering and attribution question, answered across Chapters 8 and 9.

Question 4. “When capacity is constrained, who decides what runs first, and by what rule?” This is the allocation question, answered in Chapter 11.

Question 5. “Where, exactly, does this AI flow pay for itself, and who is on the hook for knowing?” This is the value boundary question, answered in Chapter 12.

These are the Founding Questions, and their essential qualification is “from its own records.” An organization can produce an answer by asking the provider, making an estimate, or repeating the business case. Each approach produces a number, but none produces knowledge the organization holds and can check.

A reader who has finished Part I has no method yet for answering these questions from records. Parts II through IV provide that method: Part II establishes what is true, Part III builds the five functions, and Part IV turns them into an institution.

CRAFT SECTION

The trace procedure

The **trace procedure** guides a manager through a formal claim before relying on it. In six steps, it turns a theorem from a conclusion to accept or reject as a whole into a set of smaller claims that can each be checked.

Step 1. Locate the claim in the registry by its identifier and record its name exactly as the registry gives it.

Step 2. State its scope, then restate each condition in ordinary business language, retaining every antecedent and adding none.

Step 3. List the claims it depends on by identifier and name.

Step 4. Move one level down. Restate each dependency in one sentence and note any condition it adds that the claim above did not state.

Step 5. State what the claim establishes and, separately, what it does not.

Step 6. Name the falsifying case: a case that meets every condition but does not satisfy the conclusion.

Steps 2 and 6 do most of the work. Step 2 is where an antecedent can be lost in restatement, producing a claim the registry does not make. Step 6 tests whether the reader has understood the claim: anyone who cannot describe the case that would defeat it cannot say what it rules out.

Applied to THM-004, the procedure reveals a requirement of authority at step 4 that the theorem's wording does not show. The following trace demonstrates how.

Step 1 records THM-004, "Scaled AI Deployment Requires Cost Governance for Economic Control."

Step 2 restates the conditions. Within one deployment and its measurement boundary, the theorem applies when four conditions hold: scaling expands the places where usage can occur; measurement makes activity that costs money visible; management mechanisms can be applied to that visible activity; and total cost exceeds the price of access.

Step 3 lists the four dependencies: LEM-016, LEM-002, LEM-020, and LEM-006, one for each condition. The registry marks all four as required. Some claims also have contextual dependencies, which bear on the claim without being conditions of it. Problem P2 provides an example. Step 3 records which kind each dependency is.

Step 4 restates each dependency. LEM-016 says that moving from isolated use into repeated operations can increase the number of usage events and demand points. LEM-002 says that measured and recorded activity becomes observable enough to support analysis, attribution, monitoring, or management. LEM-020 says that monitoring, limits, attribution, or other management mechanisms can be applied to visible usage or exposure. LEM-006 says that a buyer who incurs several kinds of cost pays more in total than the access price.

Step 4 also asks what each dependency adds. LEM-020's scope requires the organization to have reliable visibility and the authority to apply management mechanisms.[17] That requirement belongs to the scope, not to the third condition. A team that can see its costs but has no authority to cap them falls outside LEM-020's scope. The lemma therefore does not support THM-004's third condition for that team, and the theorem says nothing about it. The team is in the same position as the pilot in Section 3.3, which fails the first condition. The trace has revealed a limit that the theorem's wording does not show.

Step 5 separates what the claim establishes from what a reader might infer. THM-004 establishes that cost governance is necessary for economic control of a scaled deployment. It does not establish that governance is sufficient or costless, specify the scale at which the requirement applies or the size of the exposure, or establish that a deployment has three flows.

Step 6 names the falsifying case: a scaled deployment that meets all four conditions and whose organization controls its cost without a measurement and management apparatus. Finding such a case would show the theorem to be false.

The procedure is deliberately slow: tracing a theorem takes longer than reading it and agreeing with it. That time is worth spending once for each claim a decision rests on, because the trace shows a manager exactly what supports the decision.

Chapter summary

AI Business Economics is the science of consuming AI: an ordered body of conditional claims, each stating when it applies, what it concludes, and which other claims it depends on. The claims are linked by dependency in a registry, and this book presents and cites only certified claims.

17. AI Business Economics registry, "dag.aiom, commit 9d7ee50," August 21, 2026. LEM-020's scope condition.

The registry justifies this book, but it does not organize it. The chapters follow the order in which a manager needs to learn. The registry supports each result, allowing a reader who doubts a claim to trace its support.

A trace follows a claim down through the registry, from a theorem to its supporting lemmas and propositions. It turns a conclusion into a set of smaller claims that can each be examined.

Read a formal claim in four parts: its conditions, its scope, its non-claims, and the falsifying case that would defeat it. A formal statement fixes in advance what would count as the claim being wrong.

AI Operations Management acts on the science through five functions: sourcing, planning and budgeting, metering and attribution, allocation and routing, and holding value to account at a boundary. It begins where a model becomes a service an organization consumes and borders AIOps, MLOps, FinOps, and regulatory governance, while remaining distinct from each.

The Founding Questions are the five questions the discipline exists to answer from an organization's own records. At the end of Part I, the reader has no method yet for answering them.

Key terms

AI Business Economics

The science of the business economics of consuming AI: an ordered body of conditional claims, each stating when it applies, what it concludes, and which other claims it depends on.

AI Operations Management

The management discipline that acts on AI Business Economics: the practice by which an organization buys, plans, records, assigns, allocates, and holds accountable its consumption of AI.

Registry

The structured record of every definition, assumption, and result in AI Business Economics, with each result linked to the claims that support it.

Theorem

A registry result that combines lemmas into a conclusion with direct consequences for management. The book presents each theorem it relies on in a numbered panel.

Lemma

An intermediate registry result that combines propositions, and sometimes other lemmas, to support one or more theorems.

Proposition

A single registry claim, accepted on the basis of the definitions, axioms, or earlier claims it names as support. Lemmas combine propositions.

Certified claim

A registry claim that passes the validator's fixed conditions and whose dependencies are all certified. The book presents and cites only certified claims.

Trace

Following a registry claim down through the claims it depends on, from a theorem to its lemmas and from a lemma to its propositions.

Dependency

A claim that another claim rests on. A required dependency is one the claim cannot hold without. A contextual dependency bears on the claim without being one of its conditions.

Condition

A requirement a deployment must meet for a formal claim to apply to it. A claim says nothing about a case that fails any of its conditions.

Scope

The part of a formal claim that defines which cases it addresses, such as one deployment within one boundary. A case outside that scope is outside the claim, regardless of whether it meets the conditions.

Non-claim

A conclusion a formal claim deliberately does not establish, though a reader might be tempted to draw it. Stating the non-claims prevents a claim from being cited for more than it says.

Falsifying case

A case that meets every condition of a formal claim but does not satisfy its conclusion. Finding such a case would show the claim to be wrong.

Value boundary

The boundary within which an organization judges whether a deployment pays for itself, defined by the costs and value it counts, the period it covers, and the outcomes used to judge it. LEM-021 states that any evaluation of return on AI requires such a boundary.

Founding Questions

The five questions AI Operations Management exists to answer from an organization’s own records, one for each of its five functions.

Trace procedure

The six-step method for reading a formal claim before relying on it: locate it, state its scope and restate its conditions, list its dependencies, move one level down, distinguish what it establishes from what it does not, and name the case that would defeat it.

Discussion questions and problems

DISCUSSION QUESTIONS

1 The book insists that the registry justifies the book but does not organize it. Describe what a reader would lose if the chapters instead followed the order in which the registry derives its results.

2 Of the four borders in Table 3.1, which is hardest to defend, and why? State what evidence would move that border.

3 Restate one Founding Question in the language your chief financial officer would use, without losing any of its content. Then identify which part of the question you were most tempted to drop.

4 Chapter 2 diagnosed a deployment through its three flows. Match each flow to the functions in Section 3.4 that act on it, then explain which flow draws on the most functions and why.

PROBLEMS

P1 · WORKED

A trace of LEM-020

The craft section traced a theorem. Use the same six steps to trace one of its lemmas, LEM-020, as a model for problem P2.

Step 1. LEM-020, “Visibility Enables Management Mechanisms.” Its statement reads: “If economic exposure or usage activity is visible to the relevant actor, then monitoring, limits, attribution, or management mechanisms can be applied within the relevant scope.”

Step 2. When the people responsible for a deployment can see its usage or its cost, they can apply monitoring, limits, attribution, or other management mechanisms to it.

Step 3. The lemma has eight dependencies, all required: LEM-002 and seven propositions, PROP-032, PROP-039, PROP-041, PROP-042, PROP-043, PROP-053, and PROP-085.

Step 4. LEM-002 says that recorded activity becomes observable enough to support analysis, attribution, monitoring, or management. It adds a condition: the record must be detailed enough for the relevant person to observe the activity.[18] PROP-032 says that compute use can be measured. PROP-039 says that measured compute use can inform methods for assigning cost. PROP-041 says that metering a resource or activity records one or more measurements of it, while PROP-042 says that measured activity can be distinguished by quantity. PROP-043 says that metering produces records of what was metered. PROP-053 says that monitoring can be used to observe economic exposure, adding the condition that there is exposure to observe. PROP-085 says that once usage is visible, management decisions can draw on it. Together, these claims explain why visibility makes management possible.

Step 5. LEM-020 establishes that visibility makes management mechanisms available. It does not claim that those mechanisms work automatically or that visibility alone determines the right policy or price.[19]

Step 6. The falsifying case is an organization that can see a deployment's usage and has the authority to manage it, yet cannot apply any monitoring, limit, or attribution to it.

P2 · COMPLETION

A trace with two steps left open

THM-002, "AI Adoption Cost Is Underestimated When Access Price Is Treated as Total Cost," states a result that Part II will teach. Its statement reads: "If buyer AI adoption cost includes cost categories beyond access price, and ROI evaluation requires a defined measurement boundary for cost and value, then

18. AI Business Economics registry, "dag.aiom, commit 9d7ee50," August 21, 2026. LEM-002's scope condition.

19. AI Business Economics registry, "dag.aiom, commit 9d7ee50," August 21, 2026. LEM-020's recorded exclusions.

treating access price as total adoption cost creates underestimation risk within the evaluation boundary." Steps 1, 2, 3, and 6 are worked below. Complete steps 4 and 5.

Step 1. THM-002, as named above.

Step 2. When a buyer's cost of adopting AI includes costs beyond the access price, and evaluating the return requires a boundary covering both cost and value, treating the access price as the whole cost risks understating adoption cost.

Step 3. The theorem has four dependencies: LEM-006 and LEM-021 are required, while LEM-002 and LEM-020 are contextual. LEM-006 appears in the craft section. LEM-021, "ROI Evaluation Requires Measurement Boundary," reads: "If ROI is evaluated for AI adoption, the evaluation requires a defined boundary for relevant costs, value, time period, and observed or expected outcomes." LEM-020 illustrates how a contextual dependency contributes: it bears on what a buyer can do about an understated cost once that cost is visible, but the claim that the cost is understated holds without it.[20]

Step 4. Restate each dependency in one sentence and note any condition it adds.

Step 5. State what THM-002 establishes and what it does not. Then use the treatment of LEM-020 in step 3 as a model to explain what LEM-002 contributes, given that the theorem does not depend on it as a condition.

Step 6. The falsifying case is a buyer whose adoption cost includes categories beyond the access price and who evaluates the return within a boundary covering cost and value. The buyer treats the access price as the total adoption cost, yet the cost is not understated.

P3 · INDEPENDENT

A reply to the word-games objection

A senior colleague reads THM-004 and says: "This is common sense in numbered clauses. Anyone who has run a large deployment knows you need controls. Formalizing it adds nothing." Write a one-page reply using the four parts of a formal claim from Section 3.3. Identify at least one thing the formal state-

20. AI Business Economics registry, "dag.aiom, commit 9d7ee50," August 21, 2026. THM-002's dependencies and their recorded roles.

ment allows the colleague to do that the common-sense version does not. Do not argue that the colleague is wrong about the conclusion.

Interleaving: question 4 requires Chapter 2's three flows, and the cumulative case below requires both earlier chapters.

PART I CUMULATIVE CASE

CUMULATIVE CASE · PART I

One announcement, three readings

DATED: FEBRUARY 2024. THE COMPANY'S OWN ANNOUNCEMENT, USING ITS OWN FIGURES. ONLY THAT ANNOUNCEMENT IS USED HERE.

In February 2024, the payments company Klarna announced results from the first month of an AI assistant built on OpenAI's models. The company reported that the assistant had handled 2.3 million conversations, accounting for two-thirds of its customer-service chats. It estimated that the assistant was doing the work of 700 full-time agents. Customers resolved their questions in under two minutes, compared with eleven minutes previously. Klarna estimated that the assistant would improve its profit by $40 million in 2024.[21]

Work from this announcement alone.

First, apply Chapter 1's consumption-event inventory. Identify the consumption events the announcement describes and where each is metered.

Second, apply Chapter 2's three-flow mapping. Diagnose the usage, record, and cost-and-value flows, identifying the evidence behind each diagnosis.

Third, apply each of the five Founding Questions to the announcement. For each question, state whether the public record provides an answer drawn from the organization's own records and identify the record needed to answer it.

Keep your answers, as the case returns in Chapter 6.

21. Klarna, "Klarna AI assistant handles two-thirds of customer service chats in its first month," February 27, 2024, accessed October 1, 2026. All figures in this case are the company's own, as announced.